

$$\begin{aligned}
||ds^2|| &= e^{-2\pi T||\psi||}[\eta_{\mu\nu} + \bar{h}(x)]dx^{\mu\nu} + T^2d^2\psi \\
&= [\infty]/e^{-2T\pi T||\psi||} + T^2d^2\psi \\
&\geq [\infty]/e^{-2T\pi T||\psi||} \cdot T^2d^2\psi \\
&= \frac{n}{n+1}\Gamma^n = \int e^{-x}x^{1-t}dx \\
\beta(p,q) &= \int e^{-\sin\theta\cos\theta} \int \sin\theta\cos\theta d\theta = \int \Gamma(\gamma)'dx_m \\
\int \Gamma(\gamma)'dx_m &= \int \Gamma dx_m \cdot \frac{d}{d\gamma}\Gamma \leq \int \Gamma dx_m + \frac{d}{d\gamma}\Gamma
\end{aligned}$$